

JEE MAIN REVISION MASTER: DIFFERENTIAL EQUATIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 27 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Variable Separable & Homogeneous DE	20 PYQs	Separable DE
High	Linear Differential Equations & Integrating Factor	8 PYQs	Linear DE

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Variable Separable & Homogeneous DE	$\int \frac{dy}{g(y)} = \int f(x) dx + C$, $\quad \text{For homogeneous: substitute } y = vx \implies \frac{dy}{dx} = v + x \frac{dv}{dx}$
Linear Differential Equations & Integrating Factor	$\frac{dy}{dx} + P(x)y = Q(x) \implies \text{I.F.} = e^{\int P(x) dx}$, $\quad y \cdot \text{I.F.} = \int Q(x) \cdot \text{I.F.} dx + C$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Variable Separable & Homogeneous DE (Very High Yield — 20 PYQs)

Core Principle: Classification of order and degree, separation of variables, and reduction of homogeneous equations via $y = vx$.

■ **Caution / JEE Pitfall:** Degree of a differential equation is defined ONLY when the equation is a polynomial in its derivatives (no fractional powers or $\sin(dy/dx)$).

■ **Speed Trick:** Equations of the form $\frac{dy}{dx} = f(ax + by + c)$ can be solved by substituting $z = ax + by + c$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Sum of the squares of the order and degree (if defined) of a differential equation $2y'' + (y')^2 = \sqrt{(y')^2 - 3}$ is..." [Ans: B]
- [2026 Shift] Q#2 (MCQ): " $\int xf(x)dx + (f(x)/2) = 0$, then $f(x)$ is equal to..." [Ans: C]
- [2025 Shift] Q#1 (MCQ): "If 'a' and 'b' are the order and degree respectively of the differentiable equation. $(\frac{d^2y}{dx^2})^2 + (\frac{dy}{dx})^3 = \dots$ " [Ans: N/A]

3.2 Linear Differential Equations & Integrating Factor (High Yield — 8 PYQs)

Core Principle: Standard first-order linear differential equations and integrating factor technique for $y(x)$ and $x(y)$.

■ **Caution / JEE Pitfall:** If the equation has x racydyx, remember to divide by x throughout first so the coefficient of racydyx is 1!

■ **Speed Trick:** If the equation is non-linear in y , check if it is linear in x : $(dx/dy) + P(y)x = Q(y)$ implies I.F. = $e^{\int P(y)dy}$.

Representative Source PYQs:

- **[2026 Shift] Q#3 (MCQ):** "Integrating factor of the differential equation $(1+x^2)(dy/dx)+xy=1$ is..." **[Ans: D]**
- **[2024 Shift] Q#3 (MCQ):** "The family of curves whose x and y intercepts of a tangent at any point are respectively double the x and y coordinates of that poin..." **[Ans: N/A]**
- **[2023 Shift] Q#4 (MCQ):** "If a curve passes through the point $(1,1)$ and at any point (x,y) on the curve, the product of the slope of its tangent and x coordinat..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Sum of the squares of the order and degree (if defined) of a differential equation $2y''+(y'')^2=\sqrt{(y''')-3}$ is

- (A) 3
- (B) 20 **[Correct]**
- (C) 8
- (D) 16

Correct Answer: B

Solution Analysis: The given differential equation is $2y''+(y'')^2=\sqrt{(y''')-3}$ The highest order derivative present is y''' , so the order of the differential equation is 2 To find the degree, the equation must be written as a polynomial in derivatives. Since y''' is inside a square root, we first remove the radical by squaring both sides: $(2y''+(y'')^2)^2=y'''-3$ Expanding, $4(y'')^2+4y''(y'')^2+(y'')^4=y'''-3$ Bring all terms to one side: $(y'')^4+4y''(y'')^2+4(y'')^2-y'''+3=0$ Now it is a polynomial in derivatives. The highest order deriva

Benchmark Problem 2: KCET 2026 (Q#3)

Integrating factor of the differential equation $(1+x^2)(dy/dx)+xy=1$ is

- (A) $1+x^2$
- (B) $(1/2)\log(1+x^2)$
- (C) $\frac{1}{\sqrt{1+x^2}}$
- (D) $\sqrt{1+x^2}$ **[Correct]**

Correct Answer: D

Solution Analysis: The given differential equation is $(1+x^2)(dy/dx)+xy=1$ First write it in the standard linear form $(dy/dx)+Py=Q$ Divide the whole equation by $1+x^2$: $(dy/dx)+\frac{xy}{1+x^2}=\frac{1}{1+x^2}$ So, $P=\frac{x}{1+x^2}$ For a linear differential equation, the integrating factor is $I.F.=e^{\int Pdx}$ Thus, $I.F.=e^{\int \frac{x}{1+x^2}dx}$ Now, $\int \frac{x}{1+x^2}dx=(1/2)\log(1+x^2)$ So, $I.F.=e^{(1/2)\log(1+x^2)}$ Using $e^{\log a}=a$, $I.F.=(1+x^2)^{1/2}$ Hence, $I.F.=\sqrt{1+x^2}$ So the correct option is